

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: CRYPTOGRAPHY AND NETWORK SECURITY

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1. Event Communication Security (Caesar Cipher)

Encrypted Message (Shift = 4)

Plain Text: MEET AT NOON

Each letter is shifted 4 positions forward in the alphabet.

Plain Text M E E T A T N O O N

Cipher Text Q I I X E X R S S R

Encrypted Message: QIIX EX RSSR

Decryption

The receiver shifts each encrypted letter 4 positions backward using the same key.

$Q \rightarrow M, I \rightarrow E, I \rightarrow E, X \rightarrow T, E \rightarrow A, X \rightarrow T, R \rightarrow N, S \rightarrow O, S \rightarrow O, R \rightarrow N$

Original Message: MEET AT NOON

Explanation:

The message "**MEET AT NOON**" is encrypted using the **Caesar Cipher**, a simple classical encryption technique in which each letter of the plaintext is shifted by a fixed number of positions in the alphabet. In this case, a **shift of 4** is applied, meaning every letter is moved four places forward. For example, **M becomes Q, E becomes I, T becomes X, A becomes E, N becomes R, and O becomes S**. Spaces between words are left unchanged. After applying the shift to every letter, the original message "**MEET AT NOON**" is transformed into the encrypted message "**QIIX EX RSSR**".

An authorized member who knows the encryption rule and the shift value of **4** can easily recover the original message through the **decryption** process. During decryption, each letter of the encrypted message is shifted **four positions backward** in the alphabet. For instance, **Q becomes M, I becomes E, X becomes T, E becomes A, R becomes N, and S becomes O**. By reversing the same shift used during encryption, the receiver successfully reconstructs the original message "**MEET AT NOON**". This demonstrates how the Caesar Cipher provides a simple method of securing communication, where both the sender and receiver must know the same shift key to correctly encrypt and decrypt the message.

2. Intercepted Suspicious Message

Cipher Text: GSRH RH Z HVXIVG

Original Message: THIS IS A SECRET

Justification:

- The message is encrypted using the **Atbash Cipher**, a substitution cipher in which the alphabet is reversed ($A \leftrightarrow Z$, $B \leftrightarrow Y$, $C \leftrightarrow X$, etc.).
- After decryption, the message reads "THIS IS A SECRET," indicating that the sender intended to conceal the content.
- Such encrypted messages may be flagged by security systems because they can be used to hide sensitive, unauthorized, or potentially malicious communications. However, encryption alone is not evidence of malicious intent—many legitimate applications also use encryption to protect privacy and confidentiality.

Explanation:

The intercepted message "GSRH RH Z HVXIVG" strongly suggests the use of the Atbash Cipher, a classical substitution cipher in which each letter of the alphabet is replaced with its opposite counterpart. In this cipher, A is replaced by Z, B by Y, C by X, D by W, and so on until Z is replaced by A. Unlike the Caesar Cipher, the Atbash Cipher does not use a shifting key; instead, it follows a fixed reverse-alphabet substitution pattern. By applying this rule to every character in the intercepted text, the encrypted message "GSRH RH Z HVXIVG" is decoded as "THIS IS A SECRET."

The deduction is supported by several observations. First, the encrypted message contains repeating patterns such as "RH", which appears twice and maps naturally to the common English word "IS." The single-letter word "Z" is highly likely to represent the English article "A," since one-letter words in English are typically limited to "A" and "I." Additionally, the final word "HVXIVG" translates to "SECRET," a meaningful and commonly used word in secure communications. The spacing between words is also preserved, making it easier to recognize familiar English sentence structures. These repeating patterns, word lengths, and the fixed reverse-letter mapping are all characteristic features of the Atbash Cipher, confirming that the most likely original message is "THIS IS A SECRET."

3. Secure Messaging System (Vigenère Cipher)

Plain Text: HELLO

Keyword: KEY

The keyword is repeated until it matches the length of the message.

Plain Text H E L L O

Keyword K E Y K E

Cipher Text R I J V S

Encrypted Message: RIJVS

Explanation:

The company is using the **Vigenère Cipher**, a classical polyalphabetic substitution cipher that encrypts a message using a **keyword** instead of a single fixed shift. In this method, the keyword is repeated until it matches the length of the plaintext message. For the plaintext "**HELLO**" and the keyword "**KEY**," the keyword is extended to "**KEYKE**" so that each letter of the message has a corresponding keyword letter. Each character of the plaintext is then encrypted by shifting it forward in the alphabet according to the alphabetical position of the matching keyword letter. For example, **K** represents a shift of 10, **E** represents a shift of 4, and **Y** represents a shift of 24 (considering A = 0). Applying these shifts one letter at a time transforms **H** → **R**, **E** → **I**, **L** → **J**, **L** → **V**, and **O** → **S**, producing the final ciphertext "**RIJVS**."

The repeating keyword plays the most important role in the Vigenère Cipher because it determines a **different shift value for each character** in the plaintext. Unlike the Caesar Cipher, where every letter is shifted by the same amount, the Vigenère Cipher changes the shift according to the corresponding letter of the repeated keyword. Since the keyword "**KEY**" repeats as "**KEYKE**," each letter of "**HELLO**" is encrypted using a different shift value, creating a more secure ciphertext. This variation makes it much harder for an attacker to identify patterns based on letter frequencies because identical plaintext letters may encrypt into different ciphertext letters depending on their position and the matching keyword character. As a result, the Vigenère Cipher provides stronger security than simple monoalphabetic substitution ciphers by using the repeating keyword to control the encryption of every individual character.

4. Suspicious Digital Asset Investigation

Explanation:

During a digital forensic investigation, if investigators suspect that a company logo image contains hidden confidential information, they would examine the file using **steganalysis** techniques rather than relying on visual inspection alone. Since steganography hides data inside an image without noticeably changing its appearance, the image may look completely normal to the human eye.

Investigators first compare the file's size, format, and metadata with those of an original or similar image to identify any unusual changes. They then use specialized forensic tools to analyze the image at the pixel level, looking for abnormal patterns in the color values or the **Least Significant Bits (LSBs)** of pixels, which are commonly used to hide secret information. Statistical analysis, histogram comparison, noise analysis, and pixel distribution tests can reveal inconsistencies that indicate embedded data. Investigators may also extract hidden information using steganalysis software if they identify the embedding technique or algorithm used. Through these methods, they can determine whether the image has been manipulated to conceal confidential information even though it appears unchanged during normal viewing.

To strengthen the security of hidden communications against detection or extraction, two important techniques can be employed. The first technique is to **encrypt the secret message before embedding it into the image** using a strong encryption algorithm such as AES. Even if an attacker successfully detects and extracts the hidden data, the information will remain unreadable without the correct encryption key, providing an additional layer of protection. The second technique is to use **adaptive or randomized steganography**, where the hidden data is embedded in complex or textured regions of the image instead of sequentially modifying pixels. Randomizing the embedding locations using a secret key significantly reduces detectable patterns and makes statistical steganalysis much more difficult. By combining strong encryption with advanced steganographic embedding methods, confidential information becomes far more resistant to detection, extraction, and unauthorized access, thereby greatly improving the overall security of hidden communications.

5. Legacy Encryption in Corporate Communication

a) Original Message

Cipher Text: WECRLTEERDSOEFEAOCAIVDEN

After reversing the 3-Rail Fence Cipher, the original message is:

WE ARE DISCOVERED FLEE AT ONCE

b) Why is it vulnerable?

- It only rearranges the positions of letters without changing them.
- Letter frequency remains the same, making frequency analysis possible.
- The key space is very small, allowing attackers to try all possible rail numbers.
- Modern computers can break this cipher quickly using brute-force methods.
- It does not provide sufficient security for protecting confidential corporate communication.

c) Practical Enhancement

A practical enhancement is to encrypt the message using a modern encryption algorithm such as AES (Advanced Encryption Standard) before transmission. AES provides strong confidentiality, protects against brute-force attacks, and is widely used for secure communication in modern organizations.

- Use TLS (Transport Layer Security) to encrypt emails and data transmitted over networks.
- Implement Public Key Infrastructure (PKI) with digital certificates for secure encryption and authentication.
- Enable Multi-Factor Authentication (MFA) to protect access to corporate communication systems.
- Apply strong encryption key management, including secure key generation, storage, rotation, and revocation.
- Use digital signatures to verify the authenticity and integrity of messages.
- Regularly update encryption software and security policies to address newly discovered vulnerabilities. Train employees to recognize phishing attacks and follow secure communication practices.
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- **Explanation:**

- The received ciphertext "**WECRLTEERDSOEFEAOCAIVDEN**" is a well-known example of the **Rail Fence Cipher** with **three rails (levels of arrangement)**. The Rail Fence Cipher is a classical transposition cipher in which the letters of the plaintext are written diagonally across a fixed number of rails and then read row by row to produce the ciphertext. Unlike substitution ciphers, this method does not change the letters themselves; instead, it only changes their positions. By reconstructing the zigzag pattern using three rails and placing the ciphertext characters back into their correct positions, the original plaintext can be recovered. Reversing the encryption process reveals the most probable original message as "**WE ARE DISCOVERED FLEE AT ONCE.**" This demonstrates how the transposition process rearranges the order of characters while preserving every original letter.
- Although the Rail Fence Cipher was useful as a simple encryption technique in the past, it is highly vulnerable in modern communication systems. The cipher has a very small number of possible rail configurations, allowing an attacker to try every possible arrangement in a short amount of time through brute-force analysis. Since the letters themselves are not substituted or encrypted, the original letter frequencies remain unchanged, making the ciphertext susceptible to frequency analysis and pattern recognition. Modern cryptanalysis tools can quickly reconstruct the zigzag pattern and recover the original message without requiring significant computational resources. Furthermore, the algorithm does not use a strong cryptographic key, making it incapable of providing adequate protection against present-day cyberattacks. As a result, anyone with basic knowledge of classical cryptography and access to automated analysis tools can easily decrypt messages protected only by the Rail Fence Cipher.
- A practical enhancement to improve the confidentiality of messages in this scenario is to **combine the transposition cipher with a modern encryption algorithm such as the Advanced Encryption Standard (AES)**. Before applying the Rail Fence arrangement, the plaintext should first be encrypted using AES with a secure secret key. This ensures that even if an attacker identifies and reverses the transposition pattern, the recovered data will still appear as meaningless encrypted text unless the correct AES key is available. Additionally, using strong key management practices, secure random key generation, and authenticated encryption significantly increases resistance to brute-force attacks, frequency analysis, and unauthorized access. By integrating a modern encryption algorithm with the existing transposition method, the confidentiality, integrity,